

Refresher Course in Calculus, Probability, and Statistics

Day 4: Inferential Statistics

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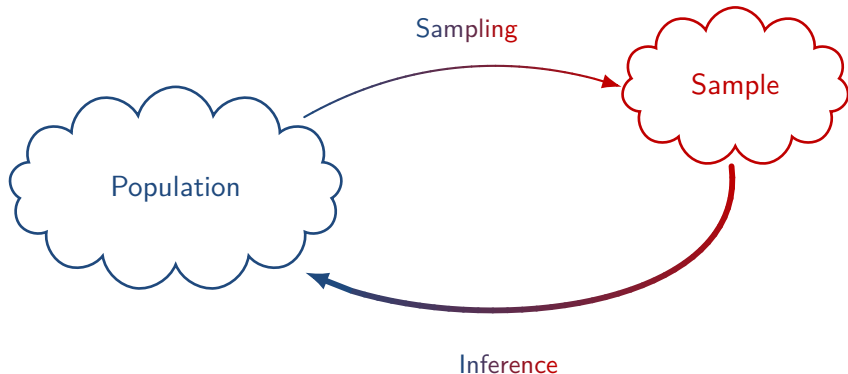
10 September 2026

Introduction

- ▶ **Inferential statistics** is the science of using data to learn about the world.
- Statistical tools help answer questions about unknown characteristics of distributions in populations of interest:
 - What is the mean earning among recent economics graduates?
 - Do earnings differ for men and women? How much?
- ▶ **Statistical inference:** learn about a population distribution from a random sample.
- Three types of statistical methods:
 - Estimation
 - Confidence intervals
 - Hypothesis testing
- Measures of association:
 - OLS

Statistical Inference

- **Definition:** use of inductive methods to reason about the distribution of a population on the basis of knowledge obtained in a sample from that population.



Random Sampling

- ▶ Samples of a population: **random sampling**.
- ▶ **Simple random sampling**:
 - each member of the population is equally likely to be included in the sample
 - the value of the random variable for the i th randomly drawn object is denoted Y_i
 - $Y_1, Y_2, \dots, Y_n$: a sample of size n from the population is a *data set*
- ▶ If the sampling is random, then:
 - two observations are drawn randomly, so the value of Y_i contains no information about the value of Y_j : $Y_1, \dots, Y_n$ are **independently** distributed
 - Y_i and Y_j are drawn from the same distribution: $Y_1, \dots, Y_n$ are **identically** distributed
 - Y_i for $i = 1, \dots, n$ are **independently and identically distributed** (i.i.d.).

Random Sampling

Population

Variable	Obs	Mean
age	2,246	39.15316
wage	2,246	7.766949

Sample 1

Var.	Obs	Mean
age	10	39.4
wage	10	6.946883

Sample 2

Var.	Obs	Mean
age	10	39.5
wage	10	8.278549

► Many different samples

→ many different values of the mean.

⇒ A whole **probability distribution** of the sample mean emerges.

Distribution of the Sample Average

- ▶ Sample average: $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.
- ▶ For a random sample, $\bar{Y}$ is a **random variable** and has a probability distribution, called the **sampling distribution**.
- ▶ Suppose $Y_1, \dots, Y_n$ are i.i.d. and the population has mean μ_Y and variance σ_Y^2 :
 - Expected value:

$$E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(Y_i) = \mu_Y.$$

- Variance:

$$\text{var}(\bar{Y}) = \frac{1}{n^2} \sum_{i=1}^n \text{var}(Y_i) + \frac{1}{n^2} \sum_{i=1}^n \sum_{j \neq i}^n \text{cov}(Y_i, Y_j) = \frac{\sigma_Y^2}{n}.$$

$$\Rightarrow \text{If } Y_i \sim N(\mu_Y, \sigma_Y^2) \Rightarrow \bar{Y} \sim N\left(\mu_Y, \frac{\sigma_Y^2}{n}\right).$$

Large-Sample Approximations to Sampling Distributions

- ▶ **Asymptotic distribution:** large-sample approximation to the sampling distribution; the approximation becomes exact as $n \rightarrow \infty$.
- ▶ **Law of large numbers:**
 - as n becomes large, $\bar{Y}$ is near μ_Y with very high probability
 - convergence in probability (consistency): $\bar{Y} \xrightarrow{p} \mu_Y$.
- ▶ **Central limit theorem:**
 - under general conditions, when n is large the distribution of $\bar{Y}$ is well approximated by a normal distribution
 - this holds *independently* of the distribution of Y
 - if Y_i ($i = 1, \dots, n$) are i.i.d. with $E(Y_i) = \mu_Y$ and $\text{var}(Y_i) = \sigma_Y^2$, then as $n \rightarrow \infty$

$$\bar{Y} \sim N\left(\mu_Y, \frac{\sigma_Y^2}{n}\right) \quad \text{or} \quad \frac{\bar{Y} - \mu_Y}{\sigma_Y/\sqrt{n}} \sim N(0, 1).$$

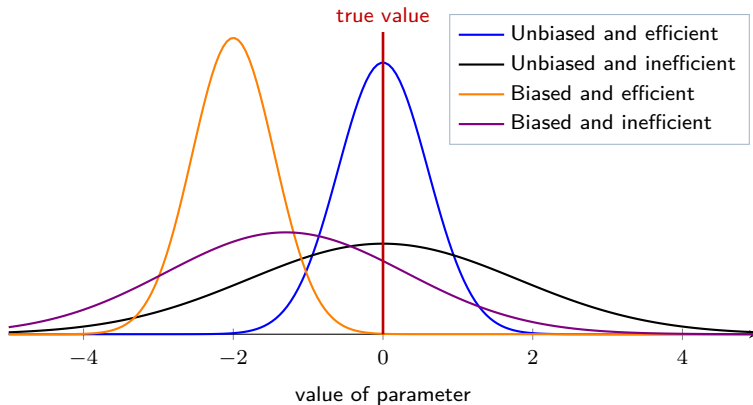
Estimators and their Properties

- ▶ **Estimator:** a function of the sample data used to infer the value of an unknown population parameter.
 - true (unknown) parameter: μ
 - estimator: $\hat{\mu}$ (read “mu hat”).
- ▶ **Estimate:** the numerical value of the estimator actually computed from a specific sample.
- ▶ Desirable characteristics of an estimator:
 - **Unbiasedness:** $E(\hat{\mu}) = \mu$ (bias = $E(\hat{\mu}) - \mu$)
 - **Consistency:** $\hat{\mu} \xrightarrow{p} \mu$ ($\text{var}(\hat{\mu}) \rightarrow 0$ as $n \rightarrow \infty$)
 - **Efficiency:** between two unbiased estimators $\hat{\mu}$ and $\tilde{\mu}$, $\hat{\mu}$ is more efficient if $\text{var}(\hat{\mu}) < \text{var}(\tilde{\mu})$.

Estimators and their Properties

(continued)

Various estimators of a population parameter



Estimation of the Population Mean

- ▶ Suppose you want to know the mean value of Y in a population
- You could estimate μ_Y exactly only with *all* values of Y
- ▶ You have at hand a sample of n i.i.d. observations $Y_1, \dots, Y_n$ with mean μ_Y and variance σ_Y^2 .
- ▶ Possible estimators:
 - Y_1
 - $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$;
 - $\tilde{Y} = \frac{1}{n} \left(\frac{1}{2}Y_1 + \frac{3}{2}Y_2 + \dots + \frac{1}{2}Y_{n-1} + \frac{3}{2}Y_n \right)$ [assume n is even].

Estimation of the Population Mean

How do these estimators fare? — Unbiasedness

How do these estimators fare judged by the three criteria? First, **unbiasedness**:

$$\blacktriangleright E(Y_1) = \mu_Y \quad \Rightarrow \text{unbiased}$$

$$\blacktriangleright E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(Y_i) = \frac{1}{n} \sum_{i=1}^n \mu_Y = \mu_Y \quad \Rightarrow \text{unbiased}$$

$$\begin{aligned} \blacktriangleright E(\tilde{Y}) &= \frac{1}{n} \left(\frac{1}{2} E(Y_1) + \frac{3}{2} E(Y_2) + \cdots + \frac{1}{2} E(Y_{n-1}) + \frac{3}{2} E(Y_n) \right) \\ &= \frac{1}{n} \left(\frac{1}{2} \cdot \frac{n}{2} \mu_Y + \frac{3}{2} \cdot \frac{n}{2} \mu_Y \right) = \mu_Y \quad \Rightarrow \text{unbiased} \end{aligned}$$

Estimation of the Population Mean

How do these estimators fare? — Efficiency

Next, **efficiency**:

$$\blacktriangleright \text{var}(Y_1) = \sigma_Y^2 > \text{var}(\bar{Y}) = \frac{\sigma_Y^2}{n} \Rightarrow \bar{Y} \text{ is more efficient than } Y_1;$$

$$\blacktriangleright \text{var}(\tilde{Y}) = \frac{5}{4} \frac{\sigma_Y^2}{n} > \text{var}(\bar{Y}) = \frac{\sigma_Y^2}{n} \Rightarrow \bar{Y} \text{ is more efficient than } \tilde{Y};$$

$$\blacktriangleright \text{var}(Y_1) = \sigma_Y^2 > \text{var}(\tilde{Y}) = \frac{5}{4} \frac{\sigma_Y^2}{n} \Rightarrow \tilde{Y} \text{ is more efficient than } Y_1.$$

$\Rightarrow \bar{Y}$ is the **most efficient** of all unbiased estimators of μ_Y , called the **Best Linear Unbiased Estimator (BLUE)**:

$\rightarrow$ for any $\hat{\mu} = \sum_{i=1}^n a_i Y_i$ with non-random constants a_i satisfying $\sum_i a_i = 1$, one has $\text{var}(\bar{Y}) \leq \text{var}(\hat{\mu})$.

Estimation of the Population Variance

► Population variance: $\text{var}(Y) = \sigma_Y^2 = E[(Y - \mu_Y)^2]$.

◦ If we knew μ_Y , we could estimate σ_Y^2 by:

$$\tilde{\sigma}_Y^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \mu_Y)^2;$$

◦ But μ_Y is unknown: it must be replaced by $\bar{Y}$.

► **Unbiased** estimator of σ_Y^2 :

$$\hat{\sigma}_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2;$$

→ Division by $n - 1$ instead of n : *degrees-of-freedom correction* as the mean uses up one degree of freedom.

► Estimator of the population **standard deviation** σ_Y : $\hat{\sigma}_Y = \sqrt{\hat{\sigma}_Y^2}$.

How Exact are Estimates of the Population Mean?

Population

$$\mu_{\text{age}} = 39.2, \quad \mu_{\text{wage}} = 7.77$$

Sample 1

Variable	Obs	Mean
age	10	39.4
wage	10	6.946883

Sample 2

Variable	Obs	Mean
age	10	39.5
wage	10	8.278549

- ▶ Because of random sampling error, it is impossible to learn the population mean μ *exactly* from a sample.
- ▶ Different samples give different estimates that scatter around μ .

Standard Error (of the Estimated Mean)

- ▶ From the CLT: as $n \rightarrow \infty$, $\bar{Y} \sim N\left(\mu_Y, \frac{\sigma_Y^2}{n}\right)$.
- ▶ Estimated variance of the estimator $\hat{\mu} = \bar{Y}$:

$$\widehat{\text{var}}(\bar{Y}) = \hat{\sigma}_{\bar{Y}}^2 = \frac{\hat{\sigma}_Y^2}{n} = \frac{1}{n} \cdot \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

- ▶ **Standard error** of $\bar{Y}$:

$$\text{se}(\bar{Y}) = \hat{\sigma}_{\bar{Y}} = \frac{\hat{\sigma}_Y}{\sqrt{n}}.$$

- estimator of the **standard deviation of the sampling distribution** of $\bar{Y}$;
→ referred to as the **standard error** of the mean / the estimate.

	Mean	Std. Err.
age	39.4	.8969083
wage	6.946883	.9061781

Confidence Intervals for the Population Mean

- **Confidence interval:** a range of values that contains μ with a specified probability, the **confidence level**.
- a 95% confidence interval for μ is constructed so that it contains the true value of μ in 95% of all possible random samples.
- Using the t -distribution ($n - 1$ degrees of freedom):

$$P\left(\bar{Y} - t_{1-\frac{\alpha}{2}, n-1} \cdot \text{se}(\bar{Y}) \leq \mu \leq \bar{Y} + t_{1-\frac{\alpha}{2}, n-1} \cdot \text{se}(\bar{Y})\right) = 1 - \alpha.$$

	Mean	Std. Err.	[95% Conf. Interval]	
age	39.4	.8969083	37.37105	41.42895
wage	6.946883	.9061781	4.896966	8.996800

Confidence Intervals for the Population Mean

(continued)

- ▶ Assuming a large sample size, the distribution is approximately normal, so we may replace the t -multiplier by the standard-normal critical value:

$$P\left(\bar{Y} - z_{\text{critical}} \cdot \text{se}(\bar{Y}) \leq \mu \leq \bar{Y} + z_{\text{critical}} \cdot \text{se}(\bar{Y})\right) = 1 - \alpha.$$

- ▶ Common two-sided critical values:
 - 90% CI: $z_{\text{critical}} = 1.645$;
 - 95% CI: $z_{\text{critical}} = 1.96$;
 - 99% CI: $z_{\text{critical}} = 2.58$.

Hypothesis Testing

- ▶ A hypothesis is a proposition about a population — a yes/no question.
- ▶ Hypotheses we might want to test:
 - Do university graduates earn CHF 6,000 per month on average?
→ hypothesis about the population mean of a *single* population;
 - Are mean earnings the same for men and women?
→ hypothesis about differences in means between *two* populations.
- ▶ Components of a hypothesis test:
 - Null hypothesis, H_0
 - Alternative hypothesis, H_1
 - Test statistic
 - Rejection region
 - Conclusion

Null and Alternative Hypotheses

- ▶ The **null hypothesis** H_0 specifies a value c for a parameter:

$$H_0 : \mu = c.$$

- H_0 is what we believe until the sample provides evidence against it; then we reject H_0 .

- ▶ **Alternative hypotheses:**

$$H_1 : \mu \neq c \quad (\text{two-sided alternative})$$

$$H_1 : \mu > c \quad (\text{one-sided alternative})$$

$$H_1 : \mu < c \quad (\text{one-sided alternative})$$

Test Statistic

- ▶ The sample information about H_0 is embodied in a **test statistic**.
- ▶ Based on the value of the test statistic, we decide whether it is reasonable to reject H_0 or not.
- ▶ Consider $H_0 : \mu = c$. If the sample comes from a population $N(\mu, \sigma)$:

$$t = \frac{\bar{Y} - c}{\hat{\sigma}/\sqrt{n}} \sim t_{n-1}.$$

- ▶ When do we use the t distribution and when the normal distribution?
 - in practice, the t *distribution* is used for *small* samples
 - when the **sample size is large**, the *normal distribution* is an accurate enough approximation.

Rejection Region

- ▶ **Rejection region:** the range of values of the t -statistic that leads to rejecting H_0 .
→ If the t -statistic falls in a region of low probability, H_0 is probably not true.
- ▶ If H_0 is false (H_1 true), the test statistic is unusually “large” or unusually “small” given the sampling distribution.

Depending on the chosen probability α , we may reject H_0 incorrectly (or fail to reject it):

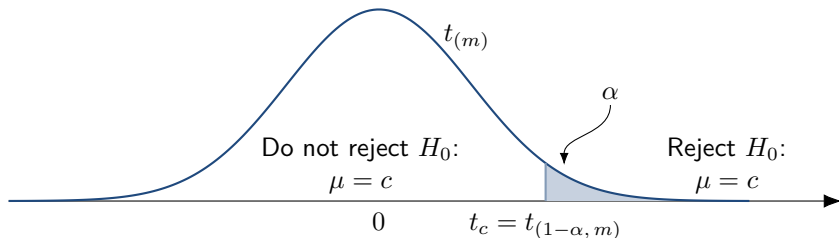
- α is the **level of significance**
- common values: $\alpha = 0.1, 0.05, 0.01$.

One-Tailed Test ($>$)

- ▶ $H_0 : \mu \leq c$ and $H_1 : \mu > c$.
- ▶ Critical value $t_{\text{critical}} = t_{(1-\alpha, n-1)}$ is the $100(1 - \alpha)$ -percentile of a t -distribution with $n - 1$ degrees of freedom:

$$P(t \leq t_{\text{critical}}) = 1 - \alpha$$

- ▶ If the test statistic $\geq t_{\text{critical}} \Rightarrow \text{reject } H_0$.



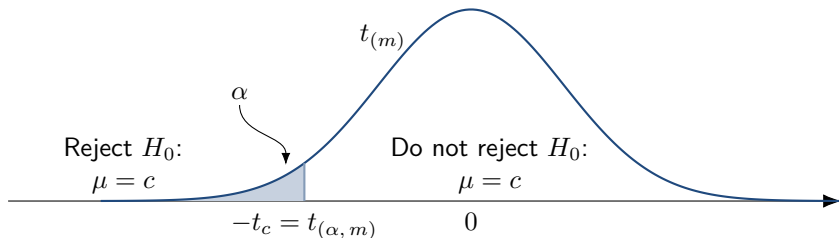
Hill, Griffiths, & Lim (2008)

One-Tailed Test ($<$)

- ▶ $H_0 : \mu \geq c$ and $H_1 : \mu < c$.
- ▶ Critical value $-t_{\text{critical}} = t_{(\alpha, n-1)}$ is the 100α -percentile of a t -distribution with $n - 1$ degrees of freedom:

$$P(t \leq -t_{\text{critical}}) = \alpha$$

- ▶ If the test statistic $\leq -t_{\text{critical}} \Rightarrow \text{reject } H_0$.

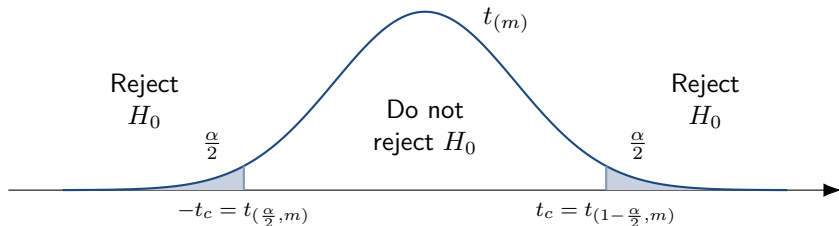


Two-Tailed Test ($\neq$)

- ▶ $H_0 : \mu = c$ and $H_1 : \mu \neq c$.
- ▶ Critical value $t_{\text{critical}} = t_{(1-\frac{\alpha}{2}, n-1)}$ is the $100(1 - \frac{\alpha}{2})$ -percentile of a t -distribution with $n - 1$ degrees of freedom:

$$P(t \geq t_{\text{critical}}) = P(t \leq -t_{\text{critical}}) = \frac{\alpha}{2}$$

- ▶ If the test statistic $\geq |t_{\text{critical}}|$ or $\leq -|t_{\text{critical}}| \Rightarrow$ reject H_0 .

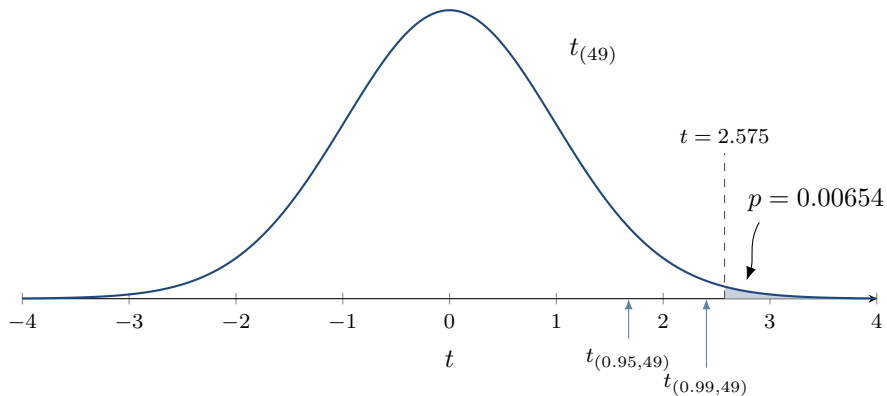


Hill, Griffiths, & Lim (2008)

The p -Value

- ▶ The p -value lets us determine the outcome of a test by comparing it to the level of significance α , without computing the critical value.
- ▶ **p -value rule:**
 - $p < \alpha \Rightarrow$ reject H_0 ;
 - $p \geq \alpha \Rightarrow$ do not reject H_0 .
- ▶ Computation of the p -value (with t the observed statistic):
 - if $H_1 : \mu < c \Rightarrow p =$ probability to the *left* of t ;
 - if $H_1 : \mu > c \Rightarrow p =$ probability to the *right* of t ;
 - if $H_1 : \mu \neq c \Rightarrow p =$ sum of the probabilities beyond $\pm|t|$ (both tails).
- ▶ When n is large (using the standard normal Φ):
 - if $H_1 : \mu < c \Rightarrow p = \Phi(t)$;
 - if $H_1 : \mu > c \Rightarrow p = 1 - \Phi(t)$;
 - if $H_1 : \mu \neq c \Rightarrow p = 2[1 - \Phi(|t|)]$.

p -Value for a One-Tailed Test ($>$)



► $t = 2.575 > t_{(0.99,49)} = 2.405$ (equivalently $p = 0.00654 < 0.01 = \alpha$):

⇒ **Reject** H_0 at the 1% level.

Conclusion of the Hypothesis Test

- ▶ When the test is completed, the conclusion is one of:
 - reject H_0 ;
 - fail to reject H_0 .
- ▶ H_0 is never said to be *accepted*: absence of evidence is not evidence of absence!
- ▶ Do not forget to interpret the test results in a meaningful way for the economic / financial problem you are working on.
- ▶ Possible mistakes:
 - **Type I error** (α)
 - False positive (of an effect / a difference)
 - H_0 rejected when it is true
 - **Type II error** (β)
 - False negative
 - H_0 not rejected when it is false

Hypothesis Test Concerning the Population Mean

- Is the mean income of US women significantly different from 10?

One-sample t test

Variable	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
wage	2,246	7.766949	.1214451	5.755523	7.528793	8.005105

`mean = mean(wage)` $t = -18.39$

H_0 : `mean = 10` `degrees of freedom = 2,245`

H_a : `mean < 10`

H_a : `mean  $\neq$  10`

H_a : `mean > 10`

$\Pr(T < t) = 0.0000$

$\Pr(|T| > |t|) = 0.0000$

$\Pr(T > t) = 1.0000$

Testing the Equality of Two Population Means

- ▶ Let two populations be distributed as $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$.
- ▶ To test the difference $\mu_1 - \mu_2$, take a random sample from each:
 - sample of size n_i from population i ;
 - sample mean $\bar{Y}_i$ and sample variance $\hat{\sigma}_i^2$.
- ▶ Null hypothesis: $H_0 : \mu_1 - \mu_2 = c$.
- ▶ **Case (a): population variances are equal.** Use both samples to estimate $\sigma_p^2 = \sigma_1^2 = \sigma_2^2$:

$$\hat{\sigma}_p^2 = \frac{(n_1 - 1)\hat{\sigma}_1^2 + (n_2 - 1)\hat{\sigma}_2^2}{n_1 + n_2 - 2}.$$

If H_0 is true:

$$t = \frac{\bar{Y}_1 - \bar{Y}_2 - c}{\sqrt{\hat{\sigma}_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \sim t_{n_1 + n_2 - 2}.$$

Testing the Equality of Two Population Means

(continued)

- **Case (b): population variances are unequal.** If H_0 is true:

$$t = \frac{\bar{Y}_1 - \bar{Y}_2 - c}{\sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_2^2}{n_2}}}.$$

- the exact distribution of this statistic is neither normal nor a standard t ;
- it is approximated by a t -distribution with degrees of freedom

$$df = \frac{\left(\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_2^2}{n_2}\right)^2}{\frac{(\hat{\sigma}_1^2/n_1)^2}{n_1 - 1} + \frac{(\hat{\sigma}_2^2/n_2)^2}{n_2 - 1}};$$

- when both n_1 and n_2 are large, the t -statistic has a standard normal distribution.

Testing the Equality of Two Population Means

Example — Hypotheses and test statistic

Question: is mean log wage the same for the two groups (white= 0 vs. white= 1)?

Summary statistics:

→ white = 0

Variable	Obs	Mean	Std. Dev.	Min	Max
ln_wage	609	1.755669	.5722596	.140951	3.707372

→ white = 1

ln_wage	1,637	1.910674	.5699247	.0049396	3.693819
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- **Step 1 — hypotheses** (difference $c = 0$; two-sided):

$$H_0 : \mu_0 - \mu_1 = 0 \text{ vs. } H_1 : \mu_0 - \mu_1 \neq 0.$$

- **Step 2 — pooled SD & standard error** (variances assumed equal):

$$\hat{\sigma}_p^2 = \frac{(n_0-1)\hat{\sigma}_0^2 + (n_1-1)\hat{\sigma}_1^2}{n_0+n_1-2} \approx 0.3256, \text{ so } \hat{\sigma}_p \approx 0.5706 \text{ and } \text{se}(\text{diff}) = \hat{\sigma}_p \sqrt{\frac{1}{n_0} + \frac{1}{n_1}} = 0.0271.$$

- **Step 3 — test statistic:** $t = \frac{\bar{Y}_0 - \bar{Y}_1 - c}{\text{se}(\text{diff})} = -\frac{0.1550}{0.0271} = -5.7237, \quad df = n_0 + n_1 - 2 = 2,244.$

Testing the Equality of Two Population Means

Example — Decision and conclusion

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Int.]	
0	609	1.755669	.0231891	.5722596	1.710128	1.801210
1	1,637	1.910674	.0140862	.5699247	1.883045	1.938303
combined	2,246	1.868645	.0121240	.5745800	1.844870	1.892420
diff		-.1550052	.0270814		-.2081123	-.1018977

$t = -5.7237$ $df = 2,244$ $\Pr(T < t) = 0.0000$ $\Pr(|T| > |t|) = 0.0000$ $\Pr(T > t) = 1.0000$

► **Step 4 — rejection region / p -value** (two-sided, $\alpha = 0.05$):

$|t| = 5.72 \gg t_{0.975, 2244} \approx 1.96$, so we reject; equivalently $p = \Pr(|T| > |t|) = 0.0000 < \alpha$ (also below 0.01). The 95% CI for the difference, $[-0.2081, -0.1019]$, does not contain 0.

- **Step 5 — conclusion:** reject H_0 . Mean log wage differs significantly between the two groups; the estimated gap is $\bar{Y}_0 - \bar{Y}_1 \approx -0.155$ log points (roughly 15% in wage terms). NB: this is a descriptive difference in group means, not a ceteris-paribus effect.